

TITLE PAGE

- Problem Statement ID – SIH26086
- Problem Statement Title - Hyperlocal Monsoon Onset & Break Prediction System
- Theme- Agriculture, FoodTech & Rural Development
- PS Category- Software
- Team ID - 173900
- Team Name - FAST&CURIOUS

Monsoon Outlook

Daily 1–4 week probabilistic monsoon outlook for every gram panchayat, turned into farmer advisories in Kannada & English.

THE PROBLEM

Farmers sow on the **first rains**. When a **false onset** or a **monsoon break** follows, seed and inputs are lost. District-level, fixed-date advisories cannot tell one panchayat from the next.

OUR SOLUTION

A **daily 1–4 week probabilistic outlook** for every **gram panchayat**: onset, false onset, dry spells, wet weeks and heavy rain. It is turned into **crop advisories** in **Kannada & English**.

WHAT THE SYSTEM DELIVERS

234

gram panchayats

Mandya district, one forecast per panchayat polygon

1–4 wk

daily outlook

issued every day from 1 May to 30 Sep

18

calibrated probabilities

onset - false onset - dry spell - dry / wet / heavy week

MJO · BSISO
ENSO · IOD

+ GEFS, GFS & our v3 AI model

ಕನ್ನಡ · English

advisory engine

web app, SMS & WhatsApp text, officer dispatch queue

Honest skill

validated on unseen seasons

the app shows what it can and cannot forecast



Farmer view in Kannada (live app)

END-TO-END PIPELINE



VALIDATION PROTOCOL



TECH STACK



100%

public data

NOAA, CHIRPS, BSISO index; IMD / NCMRWF feeds drop in

<15 s

laptop CPU per day

full outlook + advisories, 234 panchayats

<10 s

GPU per day (v3)

16-member ensemble; a full season took 15 min on one T4

Any

district

add panchayat polygons; models retrain from CHIRPS / IMD

KEY CHALLENGES AND HOW WE SOLVED THEM

Hidden look-ahead in climate indices

Switched to the real-time MJO index; onset counted only once confirmed; leak test on every input.

Coarse global models miss local rain

GEFS quantile-mapped to panchayat rain; event rules applied per ensemble member.

Over-fitting rare events

Low-capacity models, fixed selection rule, leave-one-season-out tests and an independent operational test.

Low connectivity in villages

Static, lightweight app; SMS and WhatsApp text in Kannada.

RELIABLE WHERE IT MATTERS

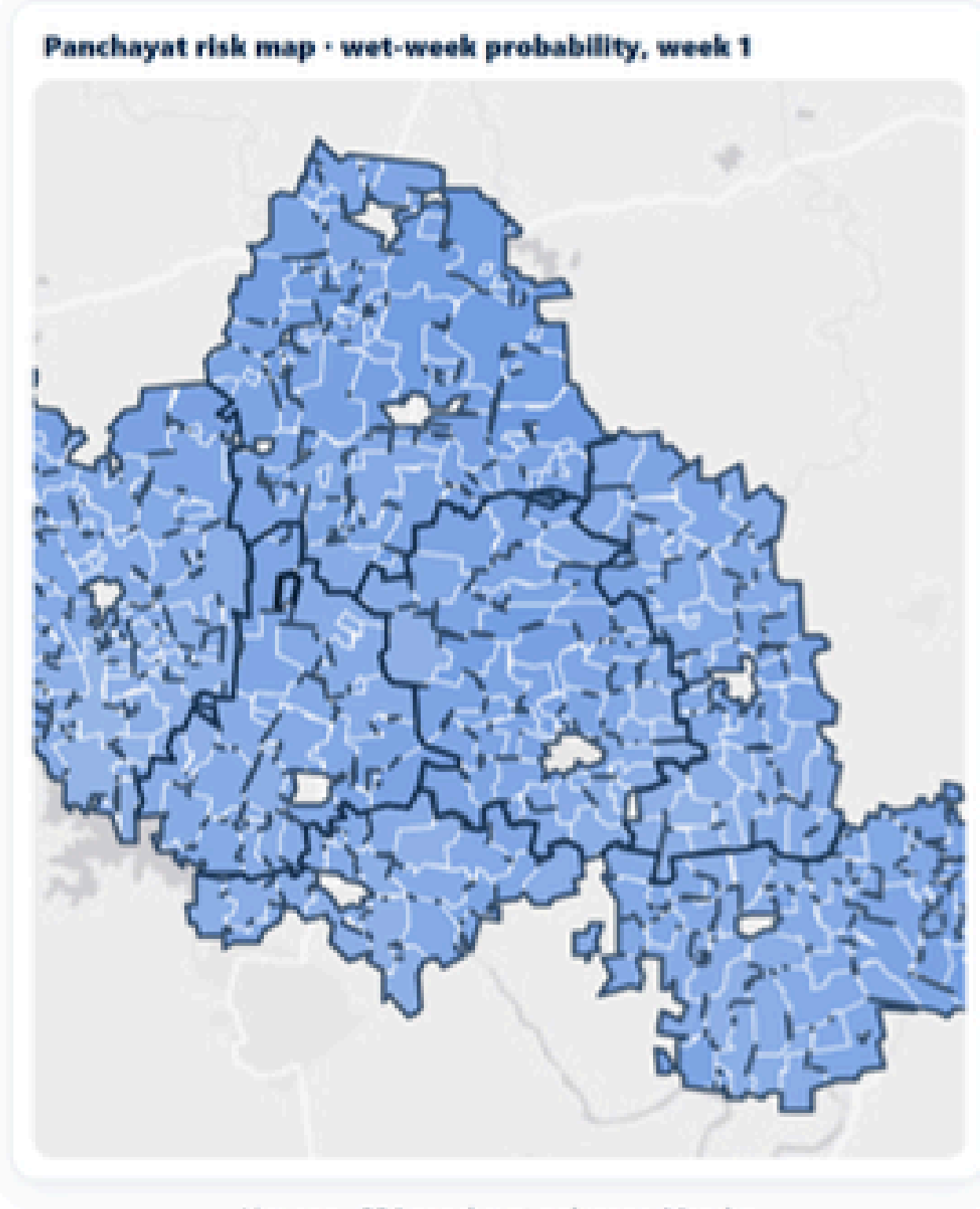
When the system says 70%, it happens about 70% of the time. Calibration gap per forecast, from the app:

FORECAST	CALIBRATION GAP
Dry week, weeks 1–4	1.2–1.9%
False-onset risk	1.4%
Wet week, week 1	4.0%
Dry spell within 3 weeks	5.6%

Limits we show openly: heavy-rain days and onset in weeks 1–3 have no reliable skill, so heavy-rain alerts are capped at amber.

VALIDATED FORECAST SKILL (vs. the long-term average, on seasons the models never saw)

Event	Week 1	Week 2	Week 3	Week 4
Dry week	+0.16	+0.15	+0.08	+0.06
Wet week	+0.23	+0.03	-	-
Dry spell (7+ days) in 3 wks	+0.08			
False onset in 3 wks	+0.06			
Heavy rain - onset week	no reliable skill yet: shown honestly, alerts capped at amber			



WHO BENEFITS

Farmers

Avoid sowing on false starts; time top-dressing and irrigation around dry spells, 1–4 weeks ahead, in Kannada.

Agriculture officers · KVKs

Block risk map and red / amber dispatch queue for targeted advisories.

IMD · NCMRWF · district

A transparent, reproducible model with reliability evidence that can be extended to other districts.

Live app · 234 panchayat polygons, Mandya

SCIENTIFIC BASIS

Real-time MJO index (ROMI)

Kiladis et al. (2014), Mon. Wea. Rev.; NOAA PSL real-time OMI

Boreal-summer ISO (BSISO) index

Kikuchi, Wang & Kajikawa (2012), Clim. Dyn.; real-time bimodal ISO index

Agronomic onset & false onset

Marteau et al. (2009); IMD dry-day threshold 2.5 mm

GEFSv12 reforecast

Hamill et al. (2022), Mon. Wea. Rev.

Diffusion downscaling (CorrDiff)

Mardani et al. (2024); basis of our v3 week-1 model

OPEN DATASETS & CODE

CHIRPS v2.0 daily rain, 0.05°

Funk et al. (2015), Sci. Data; UCSB Climate Hazards Center

NOAA GEFS / GFS forecasts

AWS Open Data (noaa-gefs-retrospective, noaa-gfs-bdp-pds); NCAR RDA ds084.1

ENSO & IOD indices

NOAA CPC weekly Niño 3.4; NOAA PSL Dipole Mode Index

Panchayat boundaries

LGD-coded gram panchayat polygons, Mandya district

Our code & full results

github.com/RohitBharadwaj-rvu/SIH-26074-downscaling-v3 · docs/results_ps26086.md